

BlueHat v17 || Dangerous Contents - Securing .Net Deserialization

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Content

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Description

Jonathan Birch, Microsoft

Serialization is a powerful tool in .Net, but if used incorrectly it can create vulnerabilities, including remote code execution. In this talk, I explain how .Net deserialization vulnerabilities occur, and why they can only be prevented by application developers. I explain four common forms of this vulnerability in detail, two using only .Net libraries and two using common vulnerable 3rd party libraries. For each of these I explain multiple ways to modify the vulnerable code to make it safe. I then use these as a basis to provide general guidelines for securing deserialization. Finally, I discuss methods for detecting .Net deserialization vulnerabilities both through static and dynamic analysis, along with coding best practices to prevent these vulnerabilities from being introduced into a product. A

handout will be provided listing potentially vulnerable API's and how to use them safely, along with useful notes on detecting this vulnerability.

Presentation Slide Deck: https://www.slideshare.net/MSbluehat/dangerous-contents-securing-net-deserialization?qid=5cf45223-06cb-4b75-b956-a5a07d1d802a&v=&b=&from_search=1

Transcript

Retrieved with yt-dlp in a container, which is the only route YouTube still answers. Expect transcription errors in type and method names.

welcome back everybody we are getting set up for Jonathan Burgess presentation on the dangers of serialization and deserialization while he is doing that we actually have a set of handouts that he prepare that should be circulating through if you do not have them right up here towards the entryway is a lovely person who will help you get a copy dotnet is a pretty magical Beast it will run damn there any code you feed it which is fantastic in terms of the versatility in the utility of the the application set the problem is that you need to be careful about what you are feeding it or you will get undesirable and often insecure outcomes Jonathan has been instrumental in helping beat the drum inside Microsoft and outside Microsoft and how to do this properly to ensure that you enjoy all of the benefits of this this broadly deployed very flexible platform without trapping yourself into an insecure deployment from which you have a hard recovery path so without further ado to give him the full amount of our time quick round of applause for Jonathan Bertrand thanks so like you said I'm Jonathan birch with Microsoft Office security and this is dangerous content securing netd serialization in this talk I'll cover how insecure use of serialization and net can lead to remote code execution vulnerabilities and then I'll tell you how you can prevent those sorts of vulnerabilities I'll give advice for specific serialization api's and I'll tell you how you can scan for serialization vulnerabilities in your products so first some background on serialization itself what is serialization well it's a process where you can take in memory objects and convert them into something like a stream or a string so that you can store them or transmit them it's usually used so that you can restore those objects either somewhere else or at a later point in time civilization is usually used to transfer data often between clients and servers is also used to backup data to a database so you can like restore program state and it lets you create equivalence between objects in different environments so JavaScript objects versus dotnet objects for instance so then how can serialization be exploited well first there's lots of serialization api's out there and many of them will do this thing where when they're deserializing a stream they'll determine what types are going to make by information from the stream itself so they're actually storing the type information in the stream and second it's often the case when serialization is used that it's being used in such a way that the stream that you're going to deserialize was at least temporarily in the hands of an untrusted party say a client user in combination this means that there's an opportunity for an attacker to inject other types in that stream that your process is going to deserialize and then you're going to create instances of types you didn't expect so why is that a problem why is it going to be bad if you make instances of types you didn't expect well first it's not going to be the case that an attacker can attack a

serial deserialization process just by creating their own malicious class and getting you to deserialize it. D serialization is only going to work if the deserializing process has knowledge of the type definition that it's going to need to make. Instead what you're going to do is an attacker is going to leverage either types in your program or types defined by the framework. Done that has a lot of types in it and a lot of them have code that will run just because you created an instance of them. Now different serialization API's work differently but many of them will call constructors on classes that they make instances of or on deserialize handlers or sometimes setters for properties they set and it doesn't matter what API you're using if you're creating an instance of a type and it has a destructor if that object gets garbage collected that destructor is going to be called. Now these methods can have side effects and it turns out that you can combine the side effects of multiple types to create gadgets that allow for things like arbitrary code execution. Here is sort of a simple example so binary formatter plus temp file collection. Binary formatter is probably the most commonly used serialization API on .NET and if you're using binary formatter to say deserialize the stream that you don't trust and a hacker has access to that stream then one way they can tack it is using the temp file collection type. Temp file collection is a type and .NET which is used to manage sort of a list of temporary files and what it will do is when it's garbage caught it will try to clean up those files and delete them so an attacker can do is if they have a stream that you're going to deserialize they can their own temp collection object add a list of files to it serialize it into the stream and when you're processed deserialize is it you'll get an object that when it gets garbage collected will delete some files in your context. Here's a diagram of that scenario so here you've got a client-server application we have like a purchase order that's being serialized by the client sent to a server where's deserialized and processed an attacker takes this system and instead serializes a temp file collection object and sends it to your server and this is sort of like a bomb it's going to go off when it gets deserialized so the server deserializes this object and then it deletes a bunch of files potentially maybe the web.config that's a good way to take down a server and this is not the best attack it's just a simple one to explain for binary formatter I encourage you to read James Forshaw's exploiting .NET manage decom presentation he outlines a way to get arbitrary remote code execution through binary formatter it's just a little more complex and I can explain in a 25-minute talk so why is this a problem well remote code execution is not something any of us want but it's also a really common issue so this vulnerability used to be present and a large number of Microsoft products we fixed every issue that we've known about if you found one please let us know it's still the case however that you can easily find web applications out on the internet that are vulnerable to this type of attack that's not something we like but we can't really go and approach every one we find on the internet who has potentially vulnerable applications so why don't we fix this instead by just making it so that doesn't have dangerous types why don't we just take all the types that have potential side effects for constructors or setters instructors and makes them so that they're safe for D serialization well it turns out that's not really practical. Done that has hundreds of thousands of types and frequently these side effects that are dangerous are going to be dangerous because of the functionality provided by the type itself we couldn't change that functionality without at the very least breaking framework compatibility and really if

you've got an application framework and you can make arbitrary in-memory objects that's probably a win condition for any attacker it's not going to be the case that you have some other ow library that you could be running on that you could do this and it probably would be okay so really the only solution to this problem is for dotnet application developers to only use serialization secure and you might be saying okay that's binary for matter but I don't use that well first you might be it turns out there's a lot of dotnet api's that use binary phone are behind the scenes so anything that reads resource files does it if you use asp.net with viewstate sort of the classic mode that uses binary formatter and I'll go back to that a bit there's various other formatter api's and net object state formatter I OS formatter other api is like net data contract serializer will do this there's a longer list in the handout and at the end of the slide deck and really even see realises that don't use binary formatter can be vulnerable so javascript serializer you would expect javascript serializer to be safe and by default it actually is by default java sterilizer won't take type information from a stream it's deserializing so you're not gonna be able to make these sort of jack-in-the-box objects that'll do bad things when the deserialized but you can use javascript sterilizer with a javascript type resolver and if you do what people usually do and set a symbol type resolver that just makes any type it's asked to this makes it similar to the binary format or scenario again however javascript serialize ER is a little bit different it will only instance types that have a public parameterless constructor and it will only set values for properties not private members like binary formatter will on the other hand it doesn't care if a type is serializable so it is exploitable but the sorts of exploits you get are different and i want to credit Alvaro Munoz and Aleksandra me Roche for publicly talking previously about JavaScript serialize their exploits like this so here's the simple payload for JavaScript serializer this is a JSON string which is just going to make an instance of an XML document object instead it's inner XML property so XML document has an a setter on its inner XML property which will just parse the XML that you give it immediately an XML document also will process document type definitions so this gives you essentially an x XC file exfiltration or server-side request forgery and you'll note for JavaScript sterilized or the deserialized call actually has a type that you pass to it that doesn't do anything if you're using a type as over at least it's only there for the case where there's no type resolver to determine what the root object type will be so that's a couple of internal or like.net defined serialization api's but there's lots of third-party serialization api's that exist also and many of those are also exploitable so there's Jason net made by Newton soft Jason done that similar to Java scripts he realized by default doesn't serial or doesn't deserialize type information from the stream but it has a type name handling property and if you set it to any value other than its default value of none you're back to the same situation where it will take type information from the stream so here's a simple gadget to get RCE from jason net it was developed by levi Broderick at Microsoft and what this is it's a JSON string that when deserialized by jason done that if typed them handling is say auto will create an instance of a Windows identity object and then set its claims identity bootstrap context property it turns out Windows identity has a non deserialize handler that will just try to use binary formatter to deserialize that bootstrap context property when it itself is deserialized and this just lets you wrap a binary formatted ow exploit inside of a jason exploit then there's service stack dot text this is an api you I looked at myself because people in

Microsoft are wanting to use it it's a little bit different it's a templated serializer so when you do a serialization or addy serialization you have to give it a type and although it puts type information in the stream it tries to make sure that the types of deserialize is match up with the ones it's expecting based off of the type you gave it however it allows for polymorphism so what will happen here is you've got a root type and the stream says well this is what the type I want you to make for that root object and it will say well is the one type assignable from the other if so it's okay otherwise throw an exception and then we'll do that for each member and all of their members and so on and it turns out that if you can walk to an object type a member of type object well that's a wild card that'll let you create any type you want so that's going to be the point where you have an exploit and in dotnet because there's a lot of innocuous seeming types that have a very large number of subclasses it's actually almost always the case that you can walk somehow through a membership graph to get to a member of type object I give an example here for a way to do it with system but exception can do with lots of other things too so once you walk to a member of type object this would let you do an exploit this is the same as the Jabra supervisor where it's an XML document where you're setting in your XML there formatting is a little bit different for what they want I did contact the maintainer zuv service talk text about this vulnerability through Microsoft vulnerability research and they did update their API to try to prevent this what they did was they added an additional whitelist which says allow list rather which says that the only types that they'll create are the ones that you add to the list or the ones that they put there by default the ones I put there by default however is every class that implements eyes serializable and dotnet which is a ridiculous number of classes I don't know a way to exploit this I don't know if there isn't one so I've outlined how different serialization api's could be exploited now I'll tell you how you can prevent those exploits so in general d serialization vulnerabilities are only going to be present with three ingredients you have to have a stream you're going to deserialize that some attacker could have modified you're going to have to be taking type information from the stream so that someone can stick a type in there you didn't expect and it can go off like a bomb like this and you're going to have to be not constraining the sort of types that it can create during the serialization if you remove any of these you're going to prevent D serialization attacks although admittedly the third one here can be a little bit tricky so for the first of those protecting the stream this is really the easiest way to prevent D serialization attacks if you only deserialize streams that you D that you serialize in the first place you're generally okay if you keep your stream on the back end that's probably all right as long as you don't have some other way for a user to modify it with like a sequel injection attack alternatively if you're going to give the user the stream to hold on to for a while like in some cases you see this where a server will give the client the session state object and it's he realized you can give them an H Mac to go with it if you validate that H Mac before doing your D serialization the user shouldn't be able to change the stream and also have the HTM X still be valid so they can't really inject their own objects the H map can also be useful for the case that you're storing a serie live stream in a database as long as you keep this secret for the hmx somewhere other than the database another thing you can do is just use a serialization API that doesn't take types from the stream so data contract serializer and xml serializable safe as far as this goes and javascript serializer and

Jason that Net are okay as long as you don't either set type name handling or add a JavaScript IPERS over there safe by default that said you can make all of these exploitable by adding some other mechanism for an attacker to determine what type will be used for the templating of the D serialization there have been cases where for instance data contract serializer was used however they added next to the data contractor data contract serializer extreme some other element that said and this is the type you used for your template when you do D serialization as soon as you do that it's exploitable again the third way you can do this and the tricky one is to contain the sorts of types that are allowed during the serialization I don't really recommend this it's very difficult to do right but it seems sometimes it's necessary for certain application designs so what you're going to do is you're gonna need to essentially create an allow list for the types that can be created during the serialization it's hard to do that correctly because there's lots of types internet that seem innocuous but are actually dangerous to deserialize hash table is an example of this you would think hash table would be safe to deserialize but it actually has a member which says this is how much memory you should allocate for the hash table and certain serialization API's will let you set that member and will then cause the hash table to try to allocate potentially arbitrary amounts of memory which lets you at least get a denial of service attack off so the way you want to do an allow list is not this if you do a typecast after an insecure serialization that's not going to actually provide you with any security most gadgets that you can get code execution through fruity serialization well I've already executed before you get an invalid cast exception for this sort of thing instead the sort of OK way to enforce an allow us for UD serialization is to use a serialization binder you can do that for binary formatter you can do that for Jason Net there's a few other serialization API's that also allow you to specify a serialization binder and the way you're going to do that is you're going to subclass a serialization binder type you're going to override the binder type method and you're going to throw an exception if that find the type method gets called with the type you didn't expect you need to keep in mind that your binder is going to be called for every type involved in the D serialization not just that root object that you're trying to deserialize so if your object has members and that up and those objects have members and so on each of those types is also going to be called or past duty or by the type method you also want to make sure that you're doing one to one type matching with this so you don't want to use is assignable from you don't know allow polymorphism that gives you the same sort of vulnerability I was explaining with service doc text you also can't just return know when you see an unexpected type you have to throw an exception because when you return null certain serialization API's will just fall back to a default binder and when that happens the default binder will just create what other type it's asked for and you don't achieve anything and you don't want to use reflection so first if you're trying to use reflection to compare type against type instead of type name against type name it's going to be really slow and you're also creating a situation where an attacker can maybe make your process load Assemblies you didn't intend to so here's an example of a good serialization binder here we're subclassing the serialization binder type we're overriding bind the type we have this list of tuples that map string the type and whenever recall with the type we make sure that it matches one of the type strings if it does we return that type if it doesn't match any of them we throw an

exception that's the way you want to do it and now I'll give you some advice on specific serialization api's so for binary formatter don't use binary formatter to deserialize an untrusted stream without a binder that is the classic D serialization vulnerability it's the one you can find every word just avoid doing it if you're using binary format or you might consider using some other serialization API instead data contract serializer and XML serialize do the job that people are trying to do with binary formatter alternatively if you're using binary for one or two C serialize a stream that you want the user to hold on to again in H Mac it's a good idea asp.net view state so asp.net view state is actually serialized in DC Ollie's using binary formatter this means that if an attacker could actually modify the view state they could get our CEO in your server which is pretty bad asp.net by default has an H Mac that it associates with that view state to prevent it from being modified the trick with this is that the secret use for the HM AK on view state is just a machine key this means that if anyone can get access to your machine key then they you can reproduce the H Mac make their own view state stick it in the query string and get RCE on your server with a get request so you need to treat the machine key for a classic asp.net server as equivalent to an admin password it needs to be difficult to guess and completely protect it however you would protect passwords Jason got net don't use Jason done that to deserialize an untrusted stream if you want to set with titanium handling set to anything other than none if you have to have type name handlings for polymorphism or something like that use a serialization binder serialization binders for json that are exactly the same as the ones for binary formatter javascript serializer don't use it with the JavaScript type resolver if you're going to actually use the armor serialize you to deserialize an untrusted stream don't allow for any method where an attack them can choose the template to type either service doc text my recommendation for service stack text is to only use it to deserialize streams that you absolutely trust if you want to use it to deserialize an untrusted stream you have to make sure that the membership graph for the object that using your template can't possibly contain a member of type object that's ridiculously difficult to do so how can you scan for serialization vulnerabilities then well first there's an obvious thing you can look for potentially dangerous serialization api's in your source code you can do this just by grep or text-based search there's the api's themselves and there's also indicators that ApS are being used in securely so if you see type name handling at all in your source it probably indicates that someone is using JSON that in an unsafe way similar if you see similarly if you see simple type resolver it's an indicator that JavaScript serialize ER or a related serialize err is being used unsafely there's a couple of tricks you can use for dynamic analysis for this too so there's a common anti pattern where binary formatter is used with a base64 encode ur and then the string that's produced is put in say oki a query string form post data things like this sometimes even in like file contents and when you have a 64 encode a binary format or serialized stream the resulting string always begins with the all-caps string a ae aad what you can do is you can look at your web traffic logs for that string if you see it it's a strong indicator that you have a potential serialization vulnerability similarly dollar sign type and underscore underscore type can indicate cases where some other serialization API is being used in such a way that the type information is taken from the stream that's going to be JavaScript serializer JSON net service doc txt so that's another thing you can look for in web traffic logs or sometimes

even in file data and that's my presentation if there's any questions [Applause] I can't possibly have explained everything everybody has ever wondered about serialization and deserialization there we go we have a question in the back hi on your sample serialization binder should you not be verifying the fully qualified name including the assembly name as part of that yes I should that was probably the most succinct answer to to a technical question I've heard him the entire he'll blue adds Bravo any additional questions for for mr. Burch what are we going to do when we get home tonight we're gonna check our serialization code please we could write a purposefully vulnerable app and have fun with it please don't do that please please don't do that so this is a set of problems that is complicated because the traditional world of identify the problem patch the problem it doesn't really it doesn't really apply in the same in the same sense here each individual person that is developing in this space needs to take accountability for what their code is doing there are tools and techniques and Jonathan's been great about showing you the ways you can start to identify these behaviors and things you might already have deployed but it at a minimum on a go-forward basis you need to be thinking about how to use serialization safely to ensure that you don't find yourself in a really messy situation further down the road so without any additional questions we have a few minutes as we reset for our next presentation a big THANK YOU to Jonathan first time [Applause]